Auditing Model Substitution in LLM APIs: Why Softwar

Commercial LLM API providers may covertly substitute cheaper models; this work shows software-based detection is unrel

Source: arxiv.org/abs/2504.04715

The Model Substitution Problem: Economic Incentives Drive Covert Swaps

Black-box APIs prevent direct verification of served weights/architecture.

Economic incentive: reduce GPU cost while preserving perceived quality.

Formalization: hypothesis test on output samples from $P(y|x)$.

Hypothesis Test

H0: $P\_actual(y|x)=P\_spec(y|x)$
H1: $P\_actual(y|x)\neq P\_spec(y|x)$

Substitution Types

1) Smaller model (e.g., 8B for 70B)
2) Quantized variant (FP8/INT8 for FP16)
3) Fine-tuned / updated version
4) Different model family

Evidence base: arxiv.org/abs/2504.04715

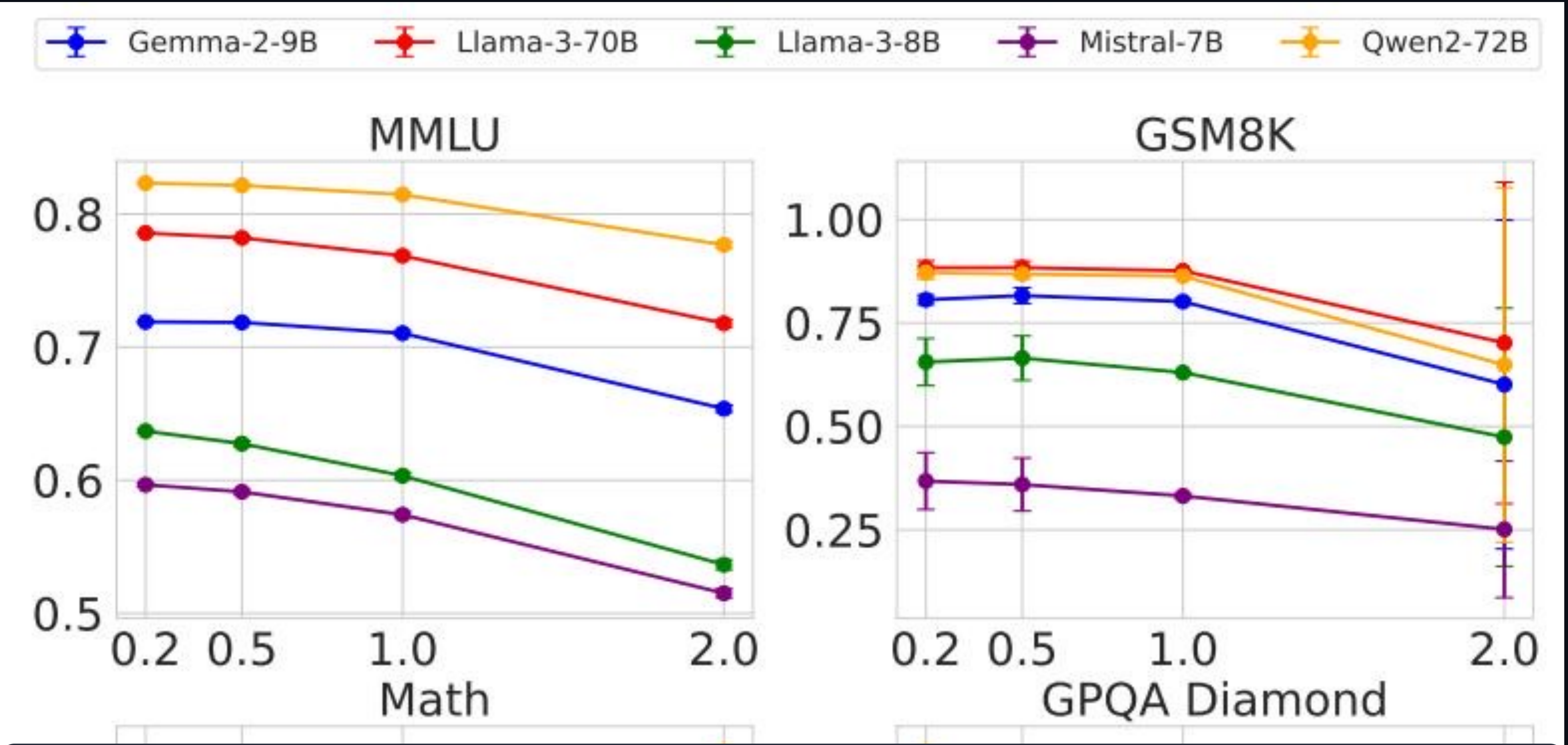
FP8 Quantization Preserves Most Accuracy — Making Substitution Hard to Detect

Model family	Variant	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B	FP16	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B	FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B	FP16	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B	FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9B	FP16	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
Gemma-2-9B	FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B	FP16	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B	FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B	FP16	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B	FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19

Interpretation

FP8 deltas are often within reported error bars (esp. MMLU/GSM8K) → substitution signal is weak.

Higher Temperature Amplifies Performance Gaps but Increases Noise



Fig

Observation

Raising temperature degrades all models; gaps can narrow or become noisy → harder to attribute differences to substitution.

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Figure

Why it breaks auditing

Production nondeterminism (software versions, kernels, GPUs) changes token log-probs—indistinguishable from substitution effects.

Randomized Routing at Low Rates Makes Statistical Detection Query-Prohibitive

Provider routes fraction p of queries to a cheaper substitute.

Observed output distribution becomes a mixture: $P_{mix} = (1-p) \cdot P_{spec} + p \cdot P_{sub}$.

As $p \to 0$, $P_{mix} \to P_{spec}$, so distinguishing requires rapidly growing query budgets.

Enables targeted evasion: route benchmark prompts to genuine model, normal traffic to substitute.

Mixture model

$$P_{mix}(y|x) = (1-p) \cdot P_{spec}(y|x) + p \cdot P_{sub}(y|x)$$

Consequence

Low-rate substitution makes software tests query-prohibitive; the auditor sees near-identical distributions.

Software-Only Auditing Methods Are Fundamentally Unreliable

Method type	Core weakness	Practical limitation
Text-based statistical tests (e.g., MMD)	Subtle substitutions (e.g., FP8) shift outputs minimally; mixtures converge to P_spec as p→0	Query budgets become impractically large; sensitive to temperature/noise
Log-probability fingerprinting	Token log-probs vary across frameworks (transformers/vLLM) and GPUs (A100/H100)	Infrastructure drift mimics substitution; hard to reproduce provider stack
Adversarial countermeasures	Provider adapts: randomized routing + benchmark evasion	Audits can be selectively passed while real traffic is substituted

Bottom line

If the provider controls the black box, software-only auditing cannot provide robust, low-cost integrity guarantees.

Sur

Trusted Execution Environments Provide Provable Cryptographic Model Integrity

Trusted Execution Environments (TEEs) provide cryptographic attestation of code + data in use.

Attestation can bind: model weights, inference binary/config, and enclave identity.

Guarantee: auditor verifies the served model is the specified model (not just statistically similar).

Practical today: NVIDIA confidential computing on H100 GPUs (modest overhead per source brief).

Contrast: ZK proofs remain computationally prohibitive for large-model inference.

Security property

Cryptographic verification of what ran (not inference from outputs)

Operational fit

Works under temperature changes, infrastructure drift, and adversarial routing.

Proposed path: TEEs (source brief)

Key Takeaways: Close the Verification Gap with Hardware Attestation

Model substitution is economically motivated and enabled by API opacity.

FP8 quantization barely moves benchmark metrics—often within error bars—so accuracy-based checks are weak.

Inference nondeterminism across software/hardware breaks log-probability fingerprinting.

Randomized routing + benchmark evasion make statistical detection query-prohibitive.

TEEs provide the actionable alternative: verifiable inference via hardware attestation.

Call to action

Evaluate LLM API integrity as a systems problem: require attestation-based guarantees, not output-only heuristics.